

FROM INNOVATION TO COMMERCIAL IMPACT



Quantum-Optimised Agentic AI Orchestration System

QUBO optimisation that assigns every task to the right AI agent, at the lowest cost, within every rule

Elijah Kim

THE PROBLEM

Agent fleets are getting expensive

15x

more tokens used by a multi-agent system than a standard chat

Anthropic engineering, 2025

40%+

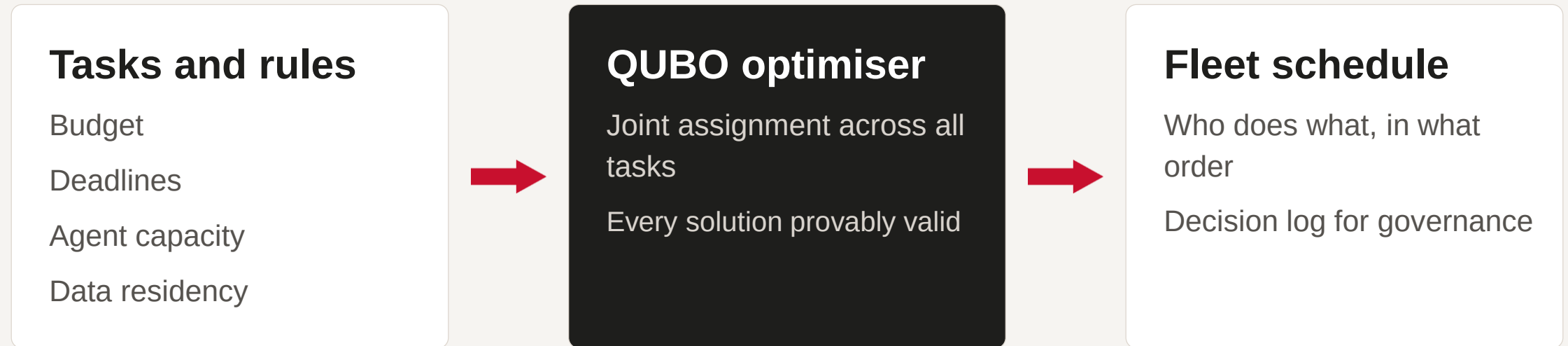
of agentic AI projects expected to be cancelled by end of 2027, citing cost, value and risk

Gartner, June 2025

Deciding which agent and model does each task is still done with fixed rules, or by asking another LLM. Neither is optimal. Neither is auditable.

OUR SOLUTION

One optimiser plans the whole fleet at once



Plugs into existing agent frameworks. Runs on classical solvers today, quantum annealing as fleets grow.

Built and tested in an MSc thesis

Proven

Penalty theorem proves every lowest-energy solution is valid

40

seeded instances across 4 fleet sizes, vs 3 classical solvers

~4×

higher success probability from calibrated penalties

2.5–3.4×

wider minimum spectral gap with our XY-mixer driver

Source: Kim, E. W. (2026). MSc thesis, Universidad Politécnica de Madrid.

OUR POSITION

Classical solvers win at today's sizes. So we ship classical now, and move the same model to quantum annealing when it pays.

Customers get savings from day one. The quantum path is an upgrade, not a dependency.

COMPETITION

Others run agents. We decide who does what.

Approach	How tasks are allocated	Fleet-wide optimum	Guaranteed rules
Agent frameworks (LangGraph, CrewAI)	Hand-written rules or prompts	No	No
Model routers	One model per prompt	No	No
Doing nothing	Premium model everywhere	No	No
Our optimiser	Joint QUBO solve	Yes	Yes

We plug into these frameworks rather than replace them.

MARKET OPPORTUNITY

A fast-growing market with a clear payer

US\$52.6B

AI agents market by 2030, up from US\$7.8B in 2025
(46.3% CAGR)

MarketsandMarkets

~US\$1.6B

avoidable inference cost a year by 2030

Our assumption: 20% inference share × 15% saving. To be tested in pilots.

Beachhead: AI consultancies and integrators in New Zealand and Australia. They feel overruns directly and deploy one tool across many clients.

Spin-out, open core, priced on savings

Pathway

Spin-out. Software improves fastest with direct customer feedback.

Channels

Open-core SDK and framework plugins, sold direct, then via cloud marketplaces.

Revenue

Subscription per workload. Enterprise tier priced as a share of savings.

Biggest uncertainty

Do benchmark savings hold on real, messy workloads? Two funded pilots measure cost, latency and quality with and without the optimiser.

What we can protect

Asset	Protection
Knowledge-store links mapped to QUBO couplings (closed loop)	Patent candidate, disclosure drafted
Penalty calibration and XY-driver design	Know-how, trade secret
Optimiser software and SDK	Copyright
Product name	Trademark, once fixed

NZ excludes computer programs “as such” (Patents Act 2013, s11), so filing would target Australia and the US, after a prior-art search and ownership confirmed with UPM and UC.

Māori data sovereignty by design

Relevance

Data residency and approved models are hard constraints, so Māori data can be guaranteed to stay onshore, with proof.

\$126B

Māori asset base in 2023, up from \$69B in 2018

Te Ōhanga Māori 2023, MBIE/BERL

We do not claim mātauranga Māori content. Next: meet a Kaiārahi Rangahau Māori in month one, then invite a Māori enterprise into a pilot on its terms.

SUCCESS IN SIX MONTHS

One pilot showing 15% or more lower inference cost at equal quality, one paying customer or letter of intent, and a clear IP position.

Ngā mihi. Thank you.